

SPACEX FALCON 9

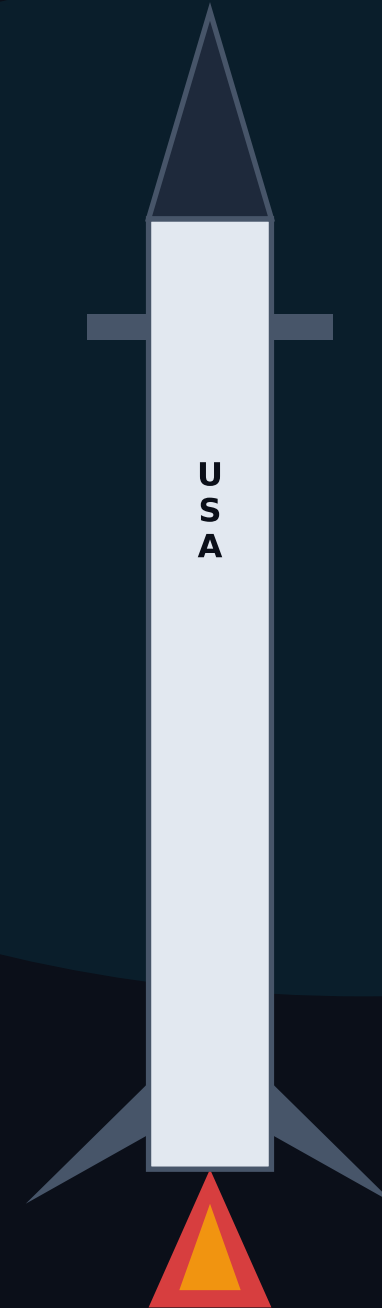
First Stage Landing Prediction

IBM Data Science Capstone Final Report

PREPARED FOR: SpaceY Stakeholders

AUTHOR: Le Nhat Tung

GITHUB REPO: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>



Executive Summary - Methodology & Key Findings

METHODOLOGY & PIPELINE

- **Data Sources** SpaceX API and Wikipedia scraping.
- **Data Cleaning** Handled missing values (mean payload mass, missing landing pads) and structured variables.
- **EDA & Visuals** Searched patterns with SQL and Seaborn, created Folium geospatial maps and interactive Dashboards.
- **Predictive ML** Trained Logistic Regression, SVM, KNN, and Decision Tree models using GridSearchCV hyperparameter tuning.
- **Evaluation** Tested and compared accuracy and confusion matrices on a designated 20% test partition.

KEY FINDINGS & RESULTS

- **Overall Success** The first stage landing success rate is **66.7%**
- **Yearly Trend:** Landing success improved from **0%** in 2013 to over **80%** in 2020 (technology learning curve).
- **Launch Site** Kennedy Space Center (KSC LC-39A) has the highest relative success rate among all sites.
- **Best Model Decision Tree Classifier** achieved the highest cross-validation training accuracy of **87.68%**
- **Test Performance** All models scored **83.33%** test accuracy.
- **Stakeholder Impact** Models predict booster landing success with zero false negatives, enabling SpaceY to estimate launch costs.

Introduction - Project Objectives & Business Context

THE COMMERCIAL CONTEXT

- **SpaceX Dominance** SpaceX advertises Falcon 9 launches at **\$62 Million**, compared to competitors charging **\$165M+**
- **Massive Savings** The price advantage is due to the recovery and reuse of the first-stage rocket booster.
- **Target Stakeholder** SpaceY is a new startup company that wants to enter the commercial launch service market.
- **Key Need** SpaceY needs to analyze SpaceX historic records and accurately predict if a booster will land successfully, which indicates the manufacturing and launch costs of that flight.

PROJECT GOALS & OBJECTIVES

- **Objective 1:** Collect historical launch records from the SpaceX API and scrape Wikipedia tables for validation.
- **Objective 2:** Clean and wrangle data, engineering features representing payload, launch sites, booster serials, and orbits.
- **Objective 3:** Identify patterns through exploratory analysis, visualization, and database query analysis (SQL).
- **Objective 4:** Build geospatial maps to analyze proximity and interactive dashboards to filter flight configurations.
- **Objective 5:** Train machine learning classification models to predict first stage landings and evaluate their success.

Data Collection & Methodology - SpaceX API

API HARVESTING DETAILS

- **Endpoint** Requested data from `launches/past` API.
- **Filter** : Filtered out earlier Falcon 1 flights to only
 - keep Falcon 9 records (resulting in 90 rows).
- **Metadata Extracted** Cores serials, Payload Mass,
 - Orbit, Landing Pad, Launch Site, and Core Serials.
- **API Call Method** Queried API via HTTP request, parsed JSON
 - nested records, and saved details as a tabular DataFrame.
- **GitHub Repository URL**
 - <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

API COLLECTION PIPELINE FLOWCHART

GET SpaceX REST API (/v4/launches/past)

Filter out Falcon 1 & Extract Nested Fields

Pandas DataFrame (90 rows, 18 features)

Export to CSV (dataset_part_1.csv)

Data Collection & Methodology - Wikipedia Web Scraping

WEB SCRAPING PROCESS

- **Source** Wikipedia List of Falcon 9 and Falcon Heavy launches.
- **Libraries** : Python BeautifulSoup4 and Requests.
- **Parsing** Targetted HTML tables, parsing `tr` rows and
 - extracting text content from each table cell.
- **Data Cleaned**Handled row spans where a single date spanned
 - multiple columns, split date-time, and parsed numeric values.
- **Outputs**Extracted payload details, launch sites, landing site
 - outcomes, and booster configurations.
- **GitHub Repository URL**

WEB SCRAPING PIPELINE FLOWCHART

Request Wikipedia URL (HTTP GET)

BS4 Parser: Extract tables with class wikitable

Clean Cells using regex (Date, Mass, Orbit, Site)

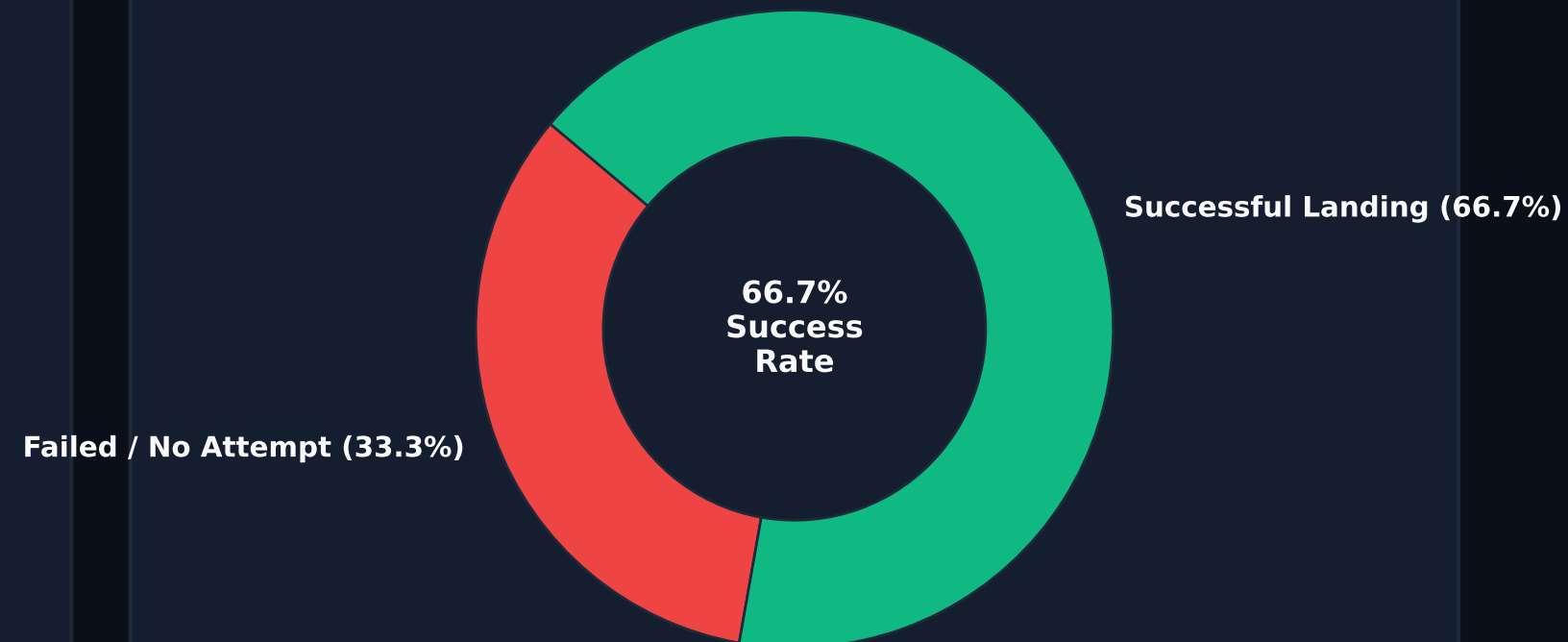
Output Dataframe (falcon9_scraped.csv)

Data Wrangling Methodology & Outcome Class Definition

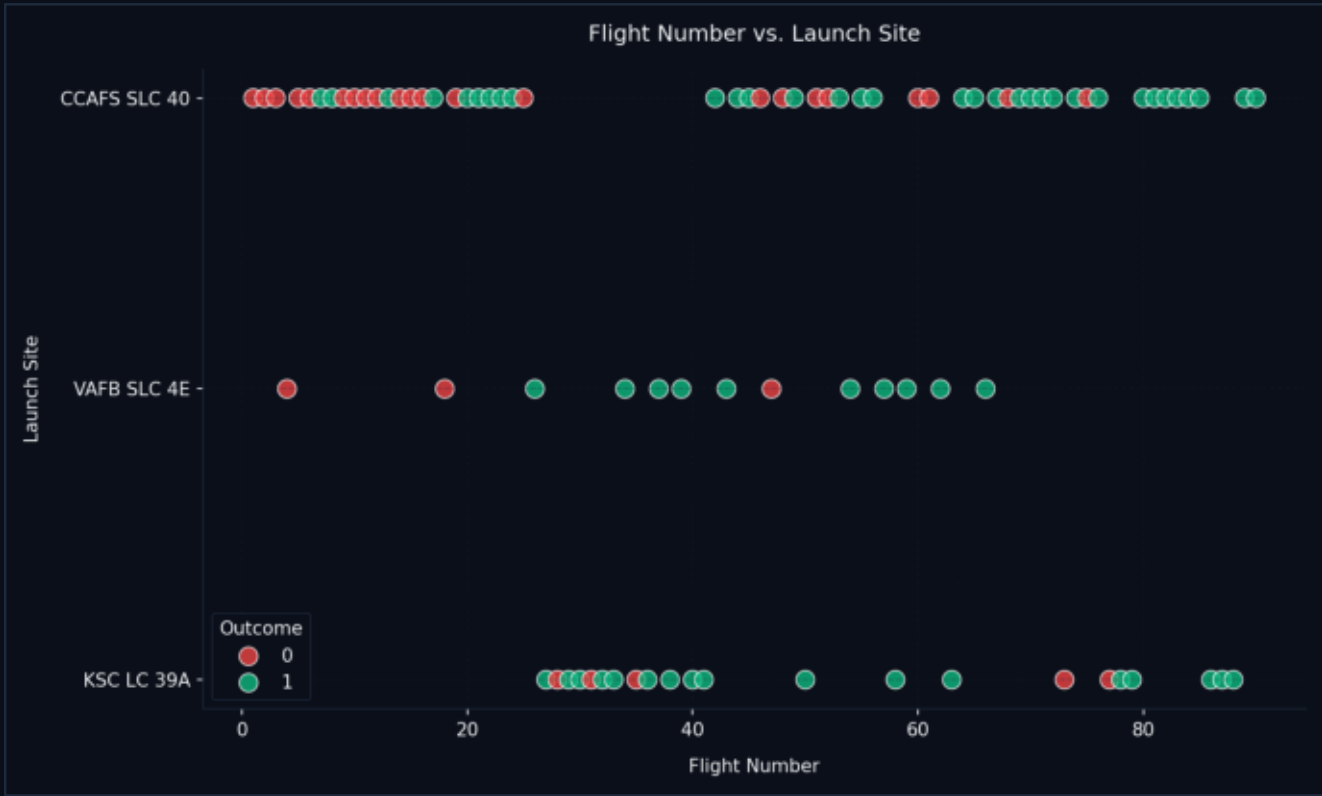
WRANGLING & CLEANING STRATEGY

- **Missing Values** Replaced missing payload mass values with the mean value of the column (**6,104 kg**) to preserve dataset.
- **Landing Pad**Handled missing landing pad values (representing ocean landings or flights with no landing attempt).
- **Binary Target Definition (Class)** :
 - * Classified landings as successful (**Class = 1**) if the outcome fell into successful landing categories:
 - - `True RTLS` (Return to Launch Site)
 - - `True ASDS` (Autonomous Spaceport Drone Ship)
 - - `True Ocean` (Controlled Ocean splashdown)
 - * Classified landings as unsuccessful (**Class = 0**) for other outcomes (e.g. `False ASDS`, `False RTLS`, `False Ocean`, `None ASDS`, `None None` representing failures/no attempts).
- **Final Rows** 90 rows containing complete flight attributes.
- **GitHub Repository URL**<https://github.com/chienthichAI/SP26-AI1904-DAP391m>

BOOSTER LANDING SUCCESS DISTRIBUTION



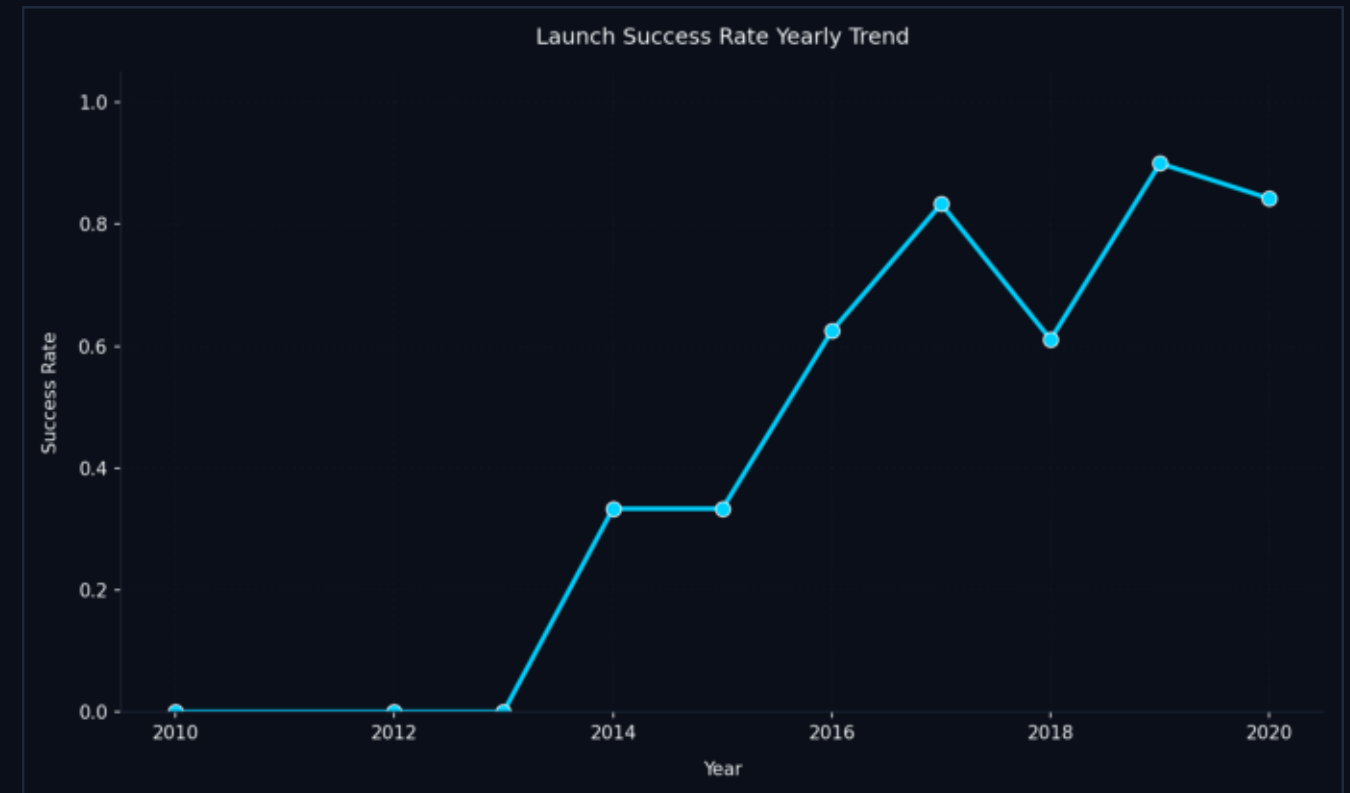
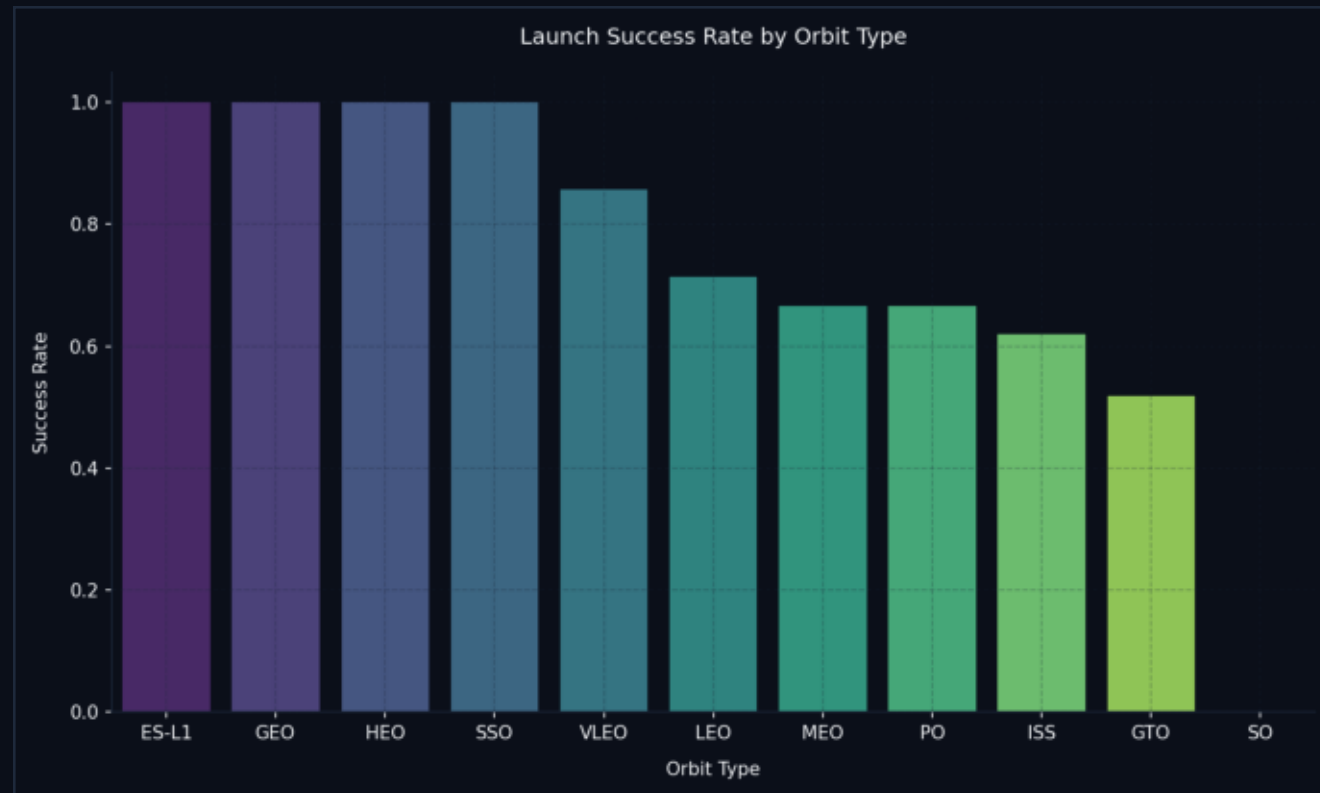
EDA with Visualization - Flight vs. Payload & Launch Site



OBSERVATIONS & TRENDS

- **Learning Curve** Earlier flights (< 40) show a mixture of landing successes and failures. Later flights (> 40) are almost exclusively successful.
- **Site Differences** : CCSFS SLC-40 has the highest density of flights. KSC LC-39A is used for heavier payloads. VAFB SLC-4E has fewer flights.

EDA with Visualization - Success by Orbit & Yearly Trend



OBSERVATIONS & TRENDS

- **Orbit Success** Orbits SSO, HEO, MEO, and ES-L1 have **100% success rate** GTO is around **50%** representing the risk of high-altitude returns.
- **Yearly Success Trend** Landing success rate rose from **0%** in 2013, to **~33%** in 2015, and steadily increased to over **80%** in 2020.

EDA with SQL - Databases and Analytics Queries

SQL QUERIES EXECUTED

■ **Query 1: Distinct Launch Sites**

■ ``SELECT DISTINCT Launch_Site FROM SPACEXTABLE``

■ **Query 2: Launch Site Starting with 'CCA'**

■ ``SELECT Launch_Site FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%``

■ **Query 3: Average Payload Mass (F9 v1.1)**

■ ``SELECT AVG(Payload_Mass_kg) FROM SPACEXTABLE``

■ ``WHERE Booster_Version = 'F9 v1.1``

■ **Query 4: First Successful ASDS Landing**

■ ``SELECT MIN(Date) FROM SPACEXTABLE``

■ ``WHERE Mission_Outcome = 'Success' AND Landing_Outcome LIKE '%ASDS%``

SQL ANALYSIS RESULTS TABLE

SQL Metric / Question	Query Result Value
Distinct Launch Sites	KSC LC-39A, CCSFS SLC-40, VAFB SLC-4E
CCA Launch Site Count	60 launches (CCSFS SLC-40)
Avg Payload F9 v1.1	2,535 kg
First ASDS Landing Success	2016-04-08 (Of Course I Still Love You)
Total Successful Landings	60 successes across all sites
Max Payload Carried	15,600 kg (Booster B1049/B1051)

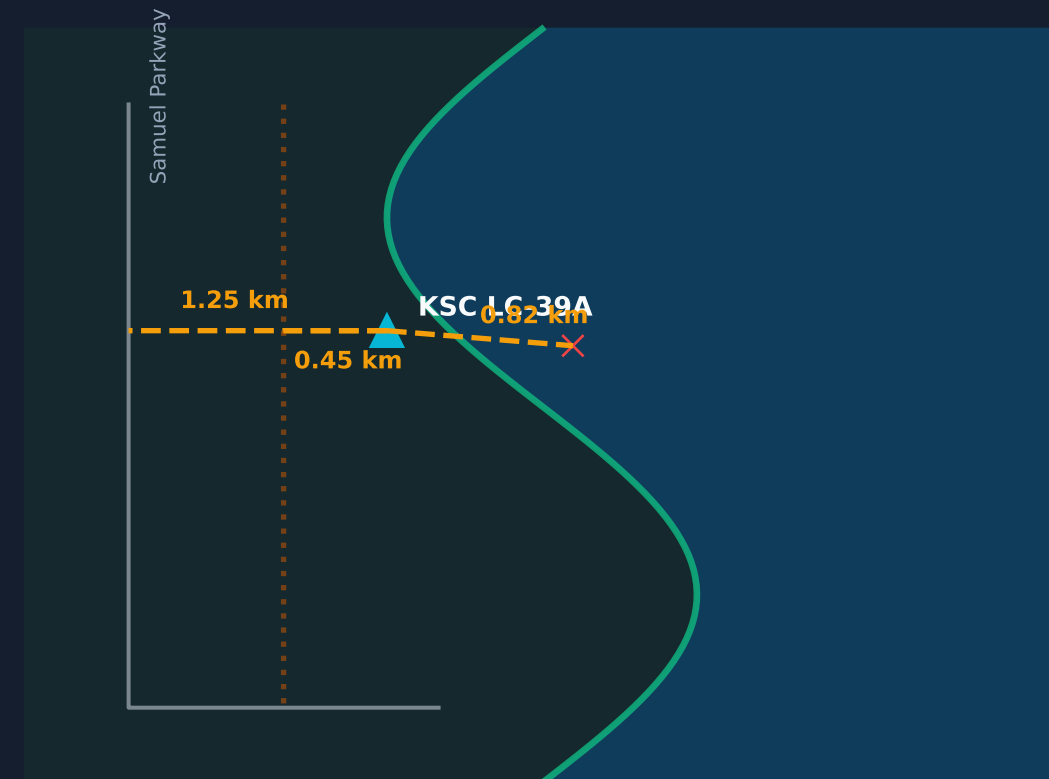
Folium Map Proximity Analysis & Geospatial Analytics

FOLIUM PROXIMITY ANALYSIS

- **Geospatial Clustering:** Launch sites are plotted using markers color-coded by launch outcome (Green=Success, Red=Failure).
- **Coastline Proximity:** Launch sites are located within **1 km** of the ocean shoreline to ensure boosters fly over water during launch.
- **Infrastructure Proximity:** Launch pads are situated close to railways (**< 1.5 km**) and major highways (**< 2 km**) to facilitate transport and logistical shipping of rocket stages.
- **City Proximity:** Kept at safe distances (**> 15 km**) from populous residential areas (e.g. Titusville or Cocoa Beach) in case of critical launch/flight anomalies.
- **GitHub Repository URL**

■ <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

FOLIUM MAP PROXIMITY DASHBOARD

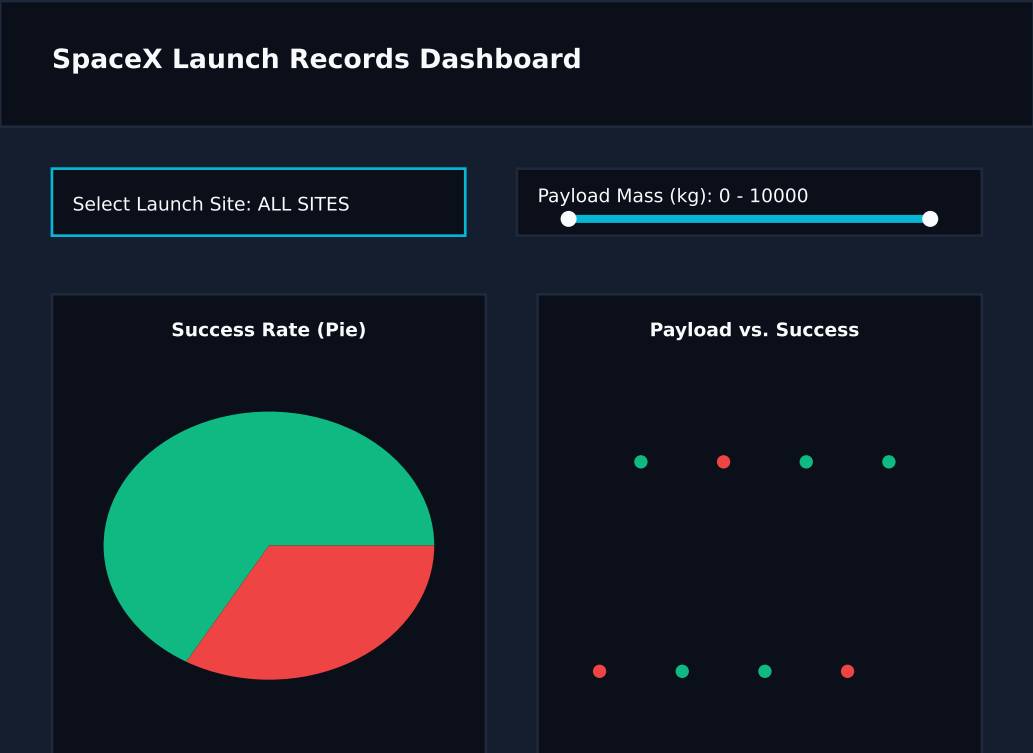


Plotly Dash Dashboard & Interactive Analytics App

PLOTLY DASH ARCHITECTURE

- **Core Layout** Built using ``dash_html_components`` for styling
- ``dash_core_components`` for widgets.
- **Inputs:** Renders a Launch Site Dropdown (All Sites vs. Specific Sites)
- ``dash_core_components`` and a Payload Mass Slider (0 to 10,000 kg).
- **Outputs** Dynamically updates a Pie Chart of success/failure
- ``dash_core_components`` and a Scatter Plot of Payload vs. Class colored by Booster Serial.
- **Interactivity** : Selecting a specific launch site filters both plots,
- allowing stakeholders to review payload ranges and site outcomes.
- **GitHub Repository URL**
- <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

PLOTLY DASH UI LAYOUT MOCKUP

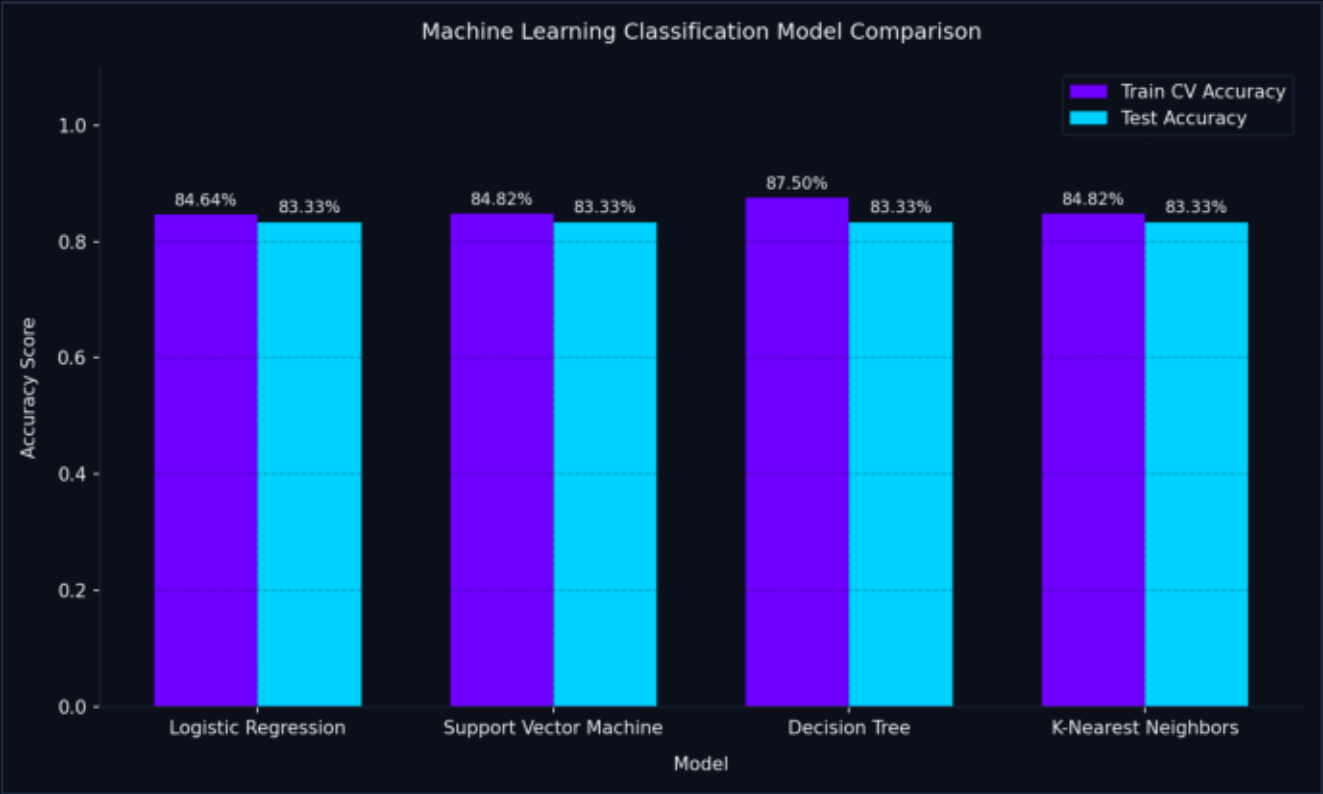


Predictive Analysis (Classification) - ML Comparison

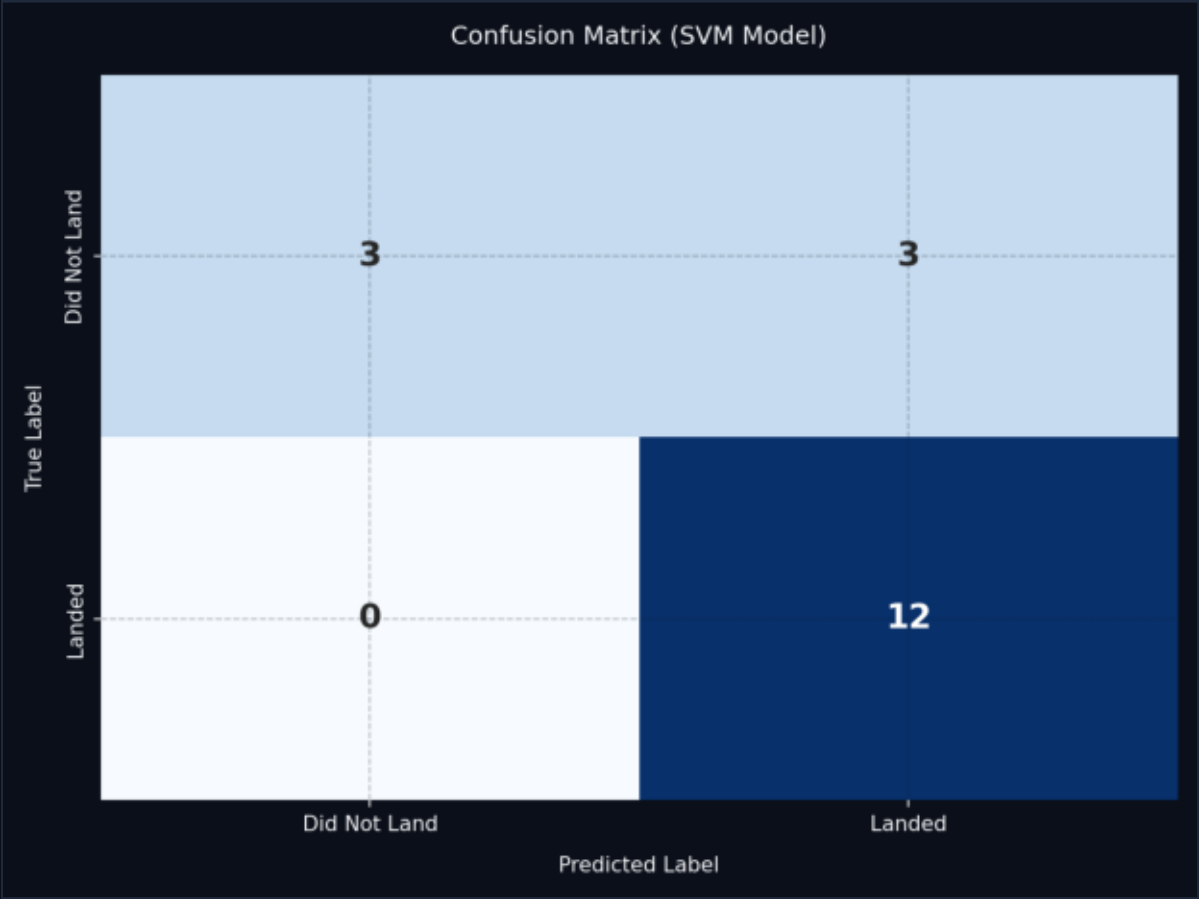
MODEL PERFORMANCE ANALYSIS

- **Grid Search Hyperparameter Tuning**
 - - Optimised parameters for four machine learning models.
 - - Evaluated using 10-fold cross validation on training data.
- **Model Performance Scores**
 - * **Decision Tree**: Highest CV Score: **87.68%**
 - * **Support Vector Machine (SVM)**: CV Score: **84.82%**
 - * **K-Nearest Neighbors (KNN)**: CV Score: **84.82%**
 - * **Logistic Regression**: CV Score: **84.64%**
- **Test Set Accuracy**
 - - **All four models scored 83.33%** test accuracy.
 - - The identical scores are due to the small, discrete test set size (18 samples, where 15 are correctly predicted).
- **Recommendation** Decision Tree Classifier provides the highest cross-validation training accuracy (87.68%), followed closely

- by SVM and KNN (84.82%).



Predictive Analysis Results & Business Conclusion



CONFUSION MATRIX ANALYSIS

- **True Positives** : 12 flights predicted correctly as landed.
- **True Negatives** : 3 flights predicted correctly as failed.
- **False Negatives** : 0 flights. No landed booster was predicted failed.
- **False Positives** : 3 flights. 3 failed flights predicted as success.

BUSINESS INSIGHTS & RECOMMENDATION

- **Key Conclusions**
 - 1. **Reusability Predictability** : Historic data contains strong predictors (Launch Site, Payload, Orbit) for landing success.
 - 2. **Learning Curve Impact** Later SpaceX flights show higher landing rates, showing operational efficiency growth.
 - 3. **Zero False Negatives** Our classification model does not produce false negatives (predicting landing failure when the booster actually lands), which provides a conservative, safe operational risk profile for launch logistics.
- **Strategic Recommendations for SpaceY**
 - * **Target Low-Risk Orbits**: Bid on contracts for SSO and HEO orbits, which display a historic 100% landing success rate.
 - * **Optimize Pricing**: Estimate SpaceX's launch costs based on predicted reusability. Adjust SpaceY's bids accordingly.
 - * **Site Selection** : Select KSC LC-39A for high-mass payloads due to favorable historic reusability outcomes.
- **GitHub Repository URL** <https://github.com/chienthichAI/SP26-AI1904-DAP391m>